

# Arnab Bhattacharjee

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## Research Interests

Human-Computer Interaction (HCI), Agentic AI, Computer Science Education (CSEd), Social Computing, Systems

## Education

**Bangladesh University of Engineering and Technology (BUET)**

Feb 2020 – Mar 2025

*B.Sc. (Honours) in Computer Science and Engineering*

CGPA: 3.88 / 4.00

## Test Scores

**IELTS Academic** (British Council, computer-delivered): 7.5 overall (Speaking 7.0)

Dec 2025

## Research Experience

**Human-AI Co-Creation for Technical Lecture Content**

Jan 2026 – Sep 2026

*HCI, Agentic AI, CSEd*

- Investigated how computer science educators author technical lecture content with generative AI tools.
- In a formative study, conducted semi-structured interviews with university educators and analyzed them using reflexive thematic analysis to identify their challenges and derive design goals.
- Guided by these goals, built an agentic pipeline that generates technical lecture content under the educator's direction.
- Conducted a qualitative user evaluation through focus group discussions (FGDs) with educators.

**Language Effects on LLM Web Agents**

Jul 2026 – Sep 2026

*Pilot research project, extension of undergraduate thesis | Supervisor: [Dr. Rifat Shahriyar](#), BUET*

- Evaluated an HTML-based LLM web agent across several widely spoken languages in two settings: a controlled offline benchmark in which the page structure stays fixed and only the language changes, and live task execution on real localized websites.
- Found that performance stayed close to English even when most of the page text was rewritten, while non-English pages consumed a substantially larger share of the model's context budget.

**Evaluating the Multilingual Capabilities of LLM-based Web UI Agent**

Nov 2023 – Mar 2025

*Undergraduate Thesis | Supervisor: [Dr. Rifat Shahriyar](#), BUET*

- Explored AI agents that complete multi-step tasks on websites for users who give instructions in languages other than English. Fine-tuned mT5-base on an extended Mind2Web dataset, benchmarked the zero-shot performance of agents built on Gemini-2.0-Flash, GPT-4o-mini, and DeepSeek-V3, and built a VLM-based automated evaluator. [Extended abstract](#).
- Poster: *How well do LLM Web Agents work for non-English Language?* LLM Agent Workshop, Carnegie Mellon University, Pittsburgh, PA, USA. May 2, 2024. [PDF](#).

## Academic and Professional Experience

**United International University (UIU)**

Jul 2025 – Present

*Lecturer, Department of Computer Science and Engineering*

*Dhaka, Bangladesh*

- Taught Data Structures and Algorithms, Object-Oriented Programming, Operating Systems, Discrete Mathematics, Digital Logic Design, Network Security, and Engineering Ethics. Coordinator of Microprocessor and Microcontroller, where student teams built projects such as a hand-gesture-controlled wheelchair ([project videos](#)).
- Trained in Outcome-Based Education (OBE), the framework of UIU CSE's Washington Accord accreditation, applied in designing course curricula and lesson plans.

**Samsung Research and Development Institute, Bangladesh**

May 2025 – Jul 2025

*Software Engineer*

*Dhaka, Bangladesh*

- Developed an Android application (Kotlin, Android Studio) for an intelligent pen that writes on any surface, with the strokes appearing on the Android device automatically.

## Projects

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- Project Tokkhok** | *React, FastAPI, PostgreSQL, Supabase, RAG* | [Demo](#) | [GitHub](#) 2025
- AI-assisted learning platform for Bengali-speaking students: chat with an uploaded knowledge base (retrieval over a vector database) in Bangla, Banglish (Bangla typed in English letters), or audio, and a smart note editor that converts Banglish notes to proper Bangla.
- Buldr-Bangladesh** | *React, Spring Boot, PostgreSQL, Tailwind CSS, Firebase Auth* | [GitHub](#) 2025
- Full-stack platform for engineers to learn (AI chatbot, smart notes, summarizing and paraphrasing tools), showcase projects, sell projects through a marketplace with order tracking, and get auto-generated portfolios.
- Machine Learning Assisted Gene Expression Restoration** | *Machine Learning, Bioinformatics* | [GitHub](#) 2024
- Explored nonlinear models for identifying the transcription factors that restore gene expression from a source cell state to a target state, extending the linear GAGER approach. Nonlinear models often reduced expression differences using fewer transcription factors across several scRNA-seq datasets.
- Deepfake Image Detector Robust Against Adversarial Attacks** | *PyTorch, ViT, YOLOv11* | [GitHub](#) 2025
- Detector for real and AI-generated images (CIFAKE). Feature dropping and augmentation reached up to 80% accuracy on adversarial data, and custom adversarial noise and dropout layers improved robustness.
- Improving the xv6 Operating System** | *C, xv6 (RISC-V)* | [GitHub](#) 2023
- Added new system calls, a multilevel feedback queue (MLFQ) scheduler, inter-process communication, and threading support to the xv6 operating system.
- Mini C Compiler** | *C, Flex, Bison, x86 Assembly* | [GitHub](#) 2023
- Built a compiler front end for a subset of C with a symbol table, lexical and semantic analysis, and intermediate code generation for x86.
- TechWorld** | *Node.js, Express.js, PostgreSQL, Oracle PL/SQL* | [GitHub](#) 2022
- E-commerce website for PCs with complex SQL and PL/SQL operations. Core feature: PC-BUILDER for assembling a custom PC from compatible parts.

## Skills

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**AI and LLMs:** Agentic LLM pipelines, RAG, LLM-as-judge evaluation, PyTorch, fine-tuning transformer models  
**HCI Methods:** Semi-structured interviews, reflexive thematic analysis, focus group discussions, formative studies

## Honors and Awards

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- Dean's List Scholarship, BUET (all four years)
- Champion, ITVerse 2023 "TechTales" Project Showcase Competition, University of Dhaka 2023
- Finalist, BUET CSE Fest Capture the Flag (CTF) Competition 2023
- Board General Scholarship, Secondary School Certificate (SSC) Examination 2017
- Talent Pool Scholarship, Junior School Certificate (JSC) Examination 2014
- Talent Pool Scholarship, Primary School Certificate (PSC) Examination 2011
- Competitive programming: 600+ problems solved, max rating 1519 on [Codeforces](#)

## References

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Available upon request.